

Machine Learning Project Proposal

Project Title

Marketing Campaign Performance Analysis

Dataset

Marketing and Product Performance Dataset

<https://www.kaggle.com/datasets/imranalishahh/marketing-and-product-performance-dataset>

Rows: 1000+

Columns: 15+

Target Variable

Revenue_Generated

Type of Task

Regression

Problem Statement

To analyze marketing campaign performance and identify factors influencing revenue generation.

Research Questions

1. Which campaigns generate the highest revenue?
2. How does budget affect revenue?
3. What is the relationship between clicks and conversions?
4. Does ROI vary across campaigns?
5. How do subscription tiers impact revenue?
6. Which features most influence revenue?
7. Can we predict revenue using machine learning?

Methodology

- * Data cleaning
- * Exploratory Data Analysis (EDA)
- * Visualization
- * Model building using Random Forest

Models Used

Random Forest Regressor

Evaluation Metrics

R² Score

Expected Outputs

- * Graphs (bar chart, scatter plot)
- * Model performance score
- * Insights on campaign performance